Geophysical Research Letters

RESEARCH LETTER
10.1029/2024GL108627

Raindrop Deformation in Turbulence

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Key Points:

- Raindrop microphysics increasingly deviate from the equilibrium state as the turbulence dissipation rate increases
- Large raindrops exhibit rounder shapes and lower fall speeds under strong turbulence
- Turbulence-invoked raindrop microphysics parameterizations are given

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Zheng, H., Zhang, Y., Li, H., Wu, Z., Xie, Y., & Zhang, L. (2024). Raindrop deformation in turbulence. *Geophysical Research Letters*, *51*, e2024GL108627. https://doi.org/10.1029/2024GL108627

Received 1 FEB 2024
Accepted 20 APR 2024

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Abstract The physical behavior of a falling raindrop is governed by delicate fluid dynamics and thermodynamics, and oscillates with time. Despite this time-variant nature, past observational and simulation studies have aimed to generalize parameterizations for describing rain microphysics bearing the assumption that raindrops fall at terminal speeds with an equilibrium shape. However, the applicability of this hypothesis in a realistic atmosphere that is inherently turbulent remains an open question. Here, we employ novel retrieval techniques to quantify the impact of turbulence on raindrop microphysics using long-term in situ observations with careful assessment of the wind effect. We find that raindrop microphysics increasingly deviate from the equilibrium state as the turbulence dissipation rate increases, and this effect is more pronounced for large raindrops. We present turbulence-invoked rain microphysical parameterizations which shed light on the complex interactions between turbulence dynamics and raindrop microphysics.

Plain Language Summary Raindrops are omnipresent in the atmosphere and play an essential role in the global hydrological cycle. The microphysics of raindrops, namely, shape, orientation and fall speed, have long been represented as a function of diameter with an equilibrium state. This simplification omits the impact of ubiquitous turbulence in the atmosphere. This study quantifies the deformation of raindrops under turbulent conditions. We find that the shape of large raindrops is more rounded and that raindrop oscillation is more pronounced in response to strong turbulence. Their fall speeds are significantly lower than the terminal speed, and the corresponding standard deviations are greater. Our work demonstrates the necessity of physically representing turbulence dynamics for accurate parameterization of raindrop microphysics via numerical simulations.

1. Introduction

Growing from tiny spherical cloud droplets, raindrops in the Earth's atmosphere are characterized by some degree of nonsphericity as well as swing (Braham, 1959). In stagnant air, the physical behaviors of raindrops are relatively stable, known as the equilibrium state, and the shape, orientation as well as fall speed can be well parameterized as a function of raindrop diameter (Beard & Chuang, 1987). This equilibrium state assumption has facilitated the parameterization of rainfall processes in multiple applications, such as remote sensing (Chandrasekar et al., 2023; Kumjian et al., 2022; Ryzhkov & Zrnic, 2019; Zheng et al., 2024), numerical models (Lin et al., 1983; Matsui et al., 2019; Morrison & Milbrandt, 2015), disaster prevention/evaluation (Li et al., 2023; Vaezi et al., 2017), etc.

In attempts to quantify the equilibrium state of raindrops, efforts have been made in recent decades from theoretical modeling (Beard, 1984; Beard & Chuang, 1987), wind tunnel experiments (Beard & Pruppacher, 1969; Pruppacher & Beard, 1970; Szakáll et al., 2009; Thurai et al., 2009), in situ observations in laboratory settings (Andsager et al., 1999; Beard & Kubesh, 1991; Beard et al., 1991; Gunn & Kinzer, 1949), and field campaigns (Huang et al., 2008; Testik & Rahman, 2016; Testik et al., 2006; Thurai & Bringi, 2005; Thurai et al., 2007, 2009). Encouragingly, a general consensus has been reached on parameterizing raindrops under calm conditions. For example, the raindrop canting angle conforms closely to a Gaussian distribution with a mean of 0° and a standard deviation of 7°–8° (Huang et al., 2008). The raindrop shape closely resembles that expected in the equilibrium state (Beard & Chuang, 1987; Thurai et al., 2007, 2009), and the raindrop fall speed is in good agreement with laboratory measurements (Atlas et al., 1973; Beard & Pruppacher, 1969; Gunn & Kinzer, 1949; Testik & Rahman, 2016). In addition, raindrop oscillation has been demonstrated (Beard et al., 1989, 2010; Szakáll et al., 2010), and the oscillation frequency is theoretically negatively correlated with raindrop diameter (Feng & Beard, 1991; Rayleigh, 1879).

Visualization: Hepeng Zheng
Writing – original draft: Hepeng Zheng, Haoran Li
Writing – review & editing: Hepeng Zheng

However, falling raindrops are subject to turbulence (Qi et al., 2022; Vela-Martín & Avila, 2021) which is ubiquitous in the atmosphere and can be enhanced via interactions between hydrometeor particles and carrier air (Bodenschatz et al., 2010; Lu et al., 2023; Zhu, Yang, et al., 2023). In particular, the air can be more turbulent in convective systems embedded in severe weather systems such as typhoons, squall lines, and supercells (Bolek & Testik, 2021; Kieu & Rotunno, 2022; Thurai et al., 2019; Zhao et al., 2021), therefore, it is necessary to understand whether and how turbulence affects raindrop microphysics. Recent case studies suggest that the observed raindrop behaviors may deviate from those at the equilibrium state (Bolek & Testik, 2021; Bringi et al., 2018; Testik & Bolek, 2023; Thurai et al., 2013, 2019). Most notably, Bolek and Testik (2021) found that rainfall microphysics change significantly during and after a tornadic storm passage. Bringi et al. (2018) and Thurai et al. (2019) suggested that reduced raindrop fall speeds are correlated with high turbulence intensities. Testik and Bolek (2023) suggested that raindrops of different sizes showed different responses to turbulence, indicating multiscale interactions between raindrop fall and turbulence.

In spite of acknowledging the inadequacy of employing raindrop parameterization derived from the equilibrium state, directly observing raindrops distorted in turbulence is challenging. Raindrops are three-dimensional (3D) objects in nature, while their characteristics are usually inferred from two-dimensional (2D) shadows measured by in situ sensors. In light of this challenge, we recently developed an algorithm to reconstruct "realistic" 3D raindrops by optimizing the information from two orthogonal 2D projections of raindrops (Zheng et al., 2023). In this study, this algorithm combined with collocated turbulence measurements was applied to more than 8 million raindrops with careful assessment of the wind effect to ascertain the impact of turbulence on raindrop canting angles, shapes and fall speeds.

2. Materials and Methods

2.1. Materials

In this study, 2D video disdrometer (2DVD) observations from two geographic regions, Jiangsu, China, and Hainan, China, were used. The climatic characteristics of the two regions are significantly different. Jiangsu is located at midlatitudes and is susceptible to midlatitude weather systems, such as fronts, midlatitude squall lines, and Western Pacific typhoons. Hainan is located in the tropics and is mainly affected by weather systems, such as tropical cyclones in the South China Sea, easterly waves and the Western Pacific Subtropical High. The locations of the two 2DVDs are labeled in Figure S1 in Supporting Information S1.

Three years of 2DVD observation data were collected from 2019 to 2021. During this time, 103 days of precipitation were observed in Hainan, with 24,626 min of valid data (accumulated precipitation: 2,050.79 mm; maximum rain rate: 144.06 mm hr^{-1}), and 39 days of precipitation were observed in Jiangsu, with 14,973 min of valid data (accumulated precipitation: 336.15 mm; maximum rain rate: 93.99 mm hr^{-1}). The corresponding information is given in Tables S1 and S2 in Supporting Information S1. The 2DVD was equipped with a 3D ultrasonic anemometer, which records wind speed and direction information every 0.1 s.

For quality control of the 2DVD data, the method proposed in a previous study (Zheng et al., 2021) was used to reject anomalous raindrops based on the empirical fall speed-diameter relationship, and raindrops with fall speeds exceeding 60% of the empirical relationship (Atlas et al., 1973) were rejected. Due to the line scan mechanism of the two orthogonal 2DVD cameras (Zhang et al., 2021), the raindrop shape contour recorded by 2DVD will be deformed when the raindrop is adverted by horizontal wind. This study followed the shape correction method proposed by Schönhuber et al. (2016). Then, the raindrop axis ratios were retrieved using the 3D oblate spheroidal reconstruction method based on the two orthogonal corrected shapes (Zheng et al., 2023). Although strong turbulence may cause raindrops to become irregular and deviate from oblate spheroidal shapes (Bolek & Testik, 2021; Thurai et al., 2019), our study still adopted the oblate spheroidal assumption, as this assumption satisfied our requirement for obtaining the core information (axis ratio) that characterizes raindrop shape. Furthermore, the relationship between the raindrop axis ratio and diameter, derived from the oblate spheroidal assumption, has significant applications in radar meteorology (Chandrasekar et al., 2023; Ryzhkov & Zrnic, 2019).

2.2. Estimation of the Turbulence Dissipation Rate

The robust second-order structure function method (Bodini et al., 2018, 2019; Muñoz-Esparza et al., 2018) was used to estimate the turbulence dissipation rate (ε) based on 3D ultrasonic anemometer data near the 2DVD. The following is a brief description of this method; please see Bodini et al. (2019) and Muñoz-Esparza et al. (2018) for additional details. Within the "inertial range", the relationship between ε and the second-order structure function can be described as follows (Kolmogorov, 1941):

$$D_U(r) \equiv <[U(x+r)-U(x)]^2> = \frac{1}{a}\varepsilon^{2/3}r^{2/3}, \tag{1}$$

where D_U is the second-order structure function, r represents the spatial separation from position x, U is the horizontal speed and $< \cdot >$ denotes an ensemble average. a is the Kolmogorov constant, which is assumed to be equal to 0.52 (Sreenivasan, 1995). The 3D ultrasonic anemometer is based on fixed-point observations; thus, Taylor's frozen turbulence hypothesis (Taylor, 1935) can be invoked to transform the spatial increment r (the position x) into the temporal increment τ (the time t) so that Equation 1 can be rewritten as follows:

$$D_U(\tau) \equiv <[U(t+\tau)-U(t)]^2> = \frac{1}{a}\varepsilon^{2/3}\left(\overline{U}\tau\right)^{2/3}, \tag{2}$$

where $\overline{U}$ represents the mean horizontal speed within a time window. Then, ε can be described as follows:

$$\varepsilon = \frac{1}{\overline{U}\tau}[aD_U(\tau)]^{3/2}. \tag{3}$$

Following the method of Bodini et al. (2019), ε is calculated every 30 s with a 2-min time window, which is centered at the time at which ε is calculated. The sampling frequency of the 3D ultrasonic anemometer is 10 Hz; that is, the wind information is recorded every 0.1 s. In every 2 min window, only the temporal increment τ between 0.1 and 2 s is used to construct the second-order structure functions for further estimation of ε. In this temporal increment range, the second-order structure functions are closer to the theoretical $\tau^{2/3}$ slopes (Bodini et al., 2019; Muñoz-Esparza et al., 2018).

3. Results and Discussion

3.1. How Turbulent Is Rainfall?

It is widely accepted that turbulence is ubiquitous in rain and can modulate microphysics. To obtain a general view of the turbulence intensity, we classified the turbulence dissipation rate (ε) into five categories at 20% intervals (Figure 1a): very weak ($\varepsilon < \varepsilon_{20\%}$), weak ($\varepsilon_{20\%} \le \varepsilon < \varepsilon_{40\%}$), moderate ($\varepsilon_{40\%} \le \varepsilon < \varepsilon_{60\%}$), strong ($\varepsilon_{60\%} \le \varepsilon < \varepsilon_{80\%}$), and very strong ($\varepsilon \ge \varepsilon_{80\%}$). The dependence of the relative fraction of various ε ranges on rainfall intensity is given in Figure 1b. For rain rates (R) greater than 50 mm hr^{-1}, the increase in R is primarily due to the increased concentration of raindrops (Wen et al., 2018). Our results show that very strong turbulence is more prevalent at heavy rain rates (34.5%), very weak turbulence is more common at weak rain rates (21.1%), and moderate turbulence is similar at both rain rates (53.5% and 60.9%). The distributions of horizontal wind speeds (U) for different ε categories are given in Figure 1c, where the turbulence increases and the distribution of U widens.

3.2. Canting Angles of Raindrops

Raindrops fall with a preferred orientation, which is known as the canting angle. Numerous studies have shown that the canting angle distribution of raindrops can be described using a Gaussian distribution with a mean canting angle of 0° and a standard deviation of σ. Figure 1d shows the impact of turbulence within σ at different ranges of raindrop equivalent volume diameters (D). It is clear that turbulence has a profound impact on the σ of larger raindrops ($D \ge 2.5$ mm): σ is lower (i.e., raindrops are more stable) under weaker turbulent conditions, while σ is higher (i.e., raindrops are more unstable) under stronger turbulent conditions. The trend of σ with increasing D under weakly turbulent conditions ($\varepsilon < \varepsilon_{40\%}$) is consistent with the results of Huang et al. (2008), who concluded that larger raindrops are more stable than smaller raindrops under calm conditions. Interestingly, our findings suggest that larger raindrops are less stable under a high ε. In addition, this finding does not depend on rainfall intensity (Figure S2 in Supporting Information S1). The potential mechanisms controlling large raindrop swing under high turbulence may be revealed by combining in situ observations and model simulations.

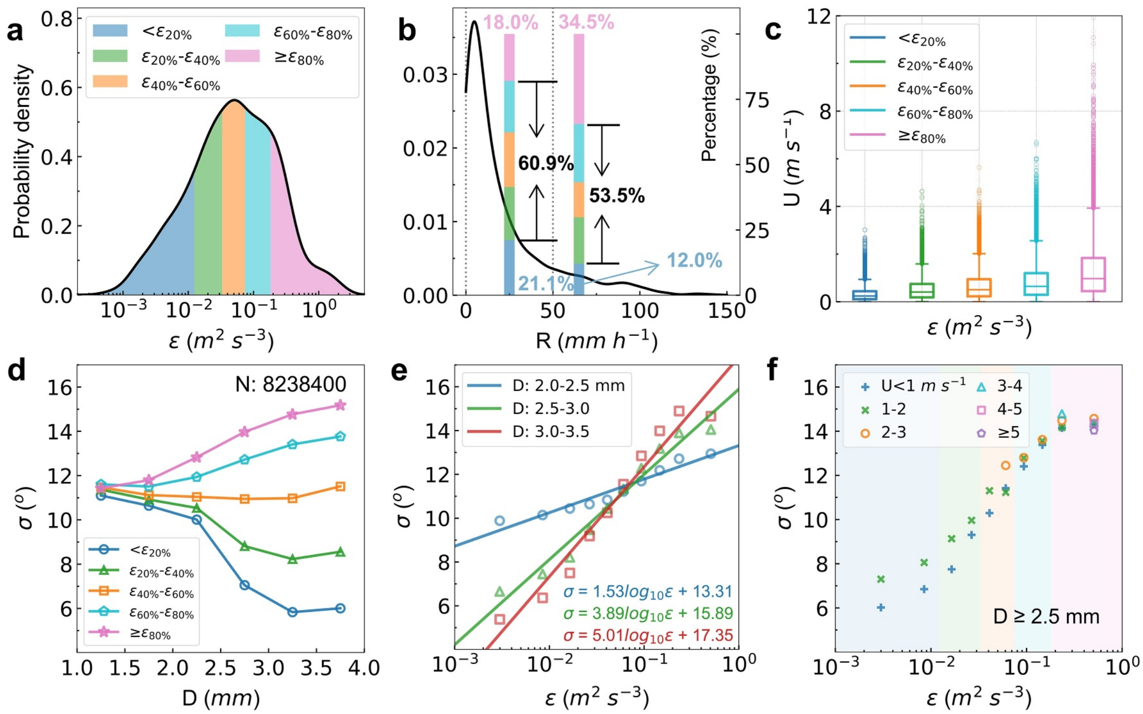

Figure 1. Effect of turbulence on the Gaussian standard deviation of the raindrop canting angle (σ). (a, b) Probability density distributions of the turbulence dissipation rate (ε; $\varepsilon_{20\%} = 0.012$ m^2 s^{-3}; $\varepsilon_{40\%} = 0.033$ m^2 s^{-3}; $\varepsilon_{60\%} = 0.075$ m^2 s^{-3}; $\varepsilon_{80\%} = 0.185$ m^2 s^{-3}) and rain rate (R). (c) Box diagram distribution of horizontal wind speeds (U) at different ε. (d) The distribution of σ for different raindrop equivalent volume diameters (D) at different ε values, where N represents the total number of raindrops. (e) Scatters and fit equations of σ with ε within different D ranges. (f) Distribution of σ with ε at different U for $D \geq 2.5$ mm (Note: only σ with a raindrop number of ≥500 is shown; ε bin: 10%).

Furthermore, we find that σ and the logarithm of ε have a good quantitative linear relationship for the same D range, as shown in Figure 1e. For D within 2.0–2.5 mm, σ varies less with increasing ε, which remains in the range of 9°–13°. For D larger than 2.5 mm, σ increases significantly with increasing ε and ranges from 5° to 15°. Therefore, the effect of turbulence on the canting angle of large raindrops deserves further attention in subsequent related studies; for example, the differential reflectivity of polarimetric radar is primarily affected by large raindrops.

3.3. Consideration of the Wind Effect

We are aware that 2DVD measurements are affected by the wind effect, which cannot be measured by a 3D ultrasonic anemometer near the 2DVD. As a result, the turbulence measured by the 3D ultrasonic anemometer near the 2DVD may differ from the turbulence in the core observation area. Using a high-speed optical disdrometer, a single-plane observation instrument without this issue, Bolek and Testik (2021) showed that there is a significant positive correlation between the standard deviation of the raindrop canting angle and turbulence intensity; that is, the stronger the turbulence near the raindrops is, the greater the standard deviation of the raindrop canting angle is. Figure 1e in this study also shows a strong positive correlation between σ and ε derived from the 3D ultrasonic anemometer near the 2DVD. This correlation allows us to assess the impact of the wind effect on our results. Furthermore, the variation in σ with horizontal wind speed (U) at different ε ranges is shown in Figure 1f. The results show that for large raindrops ($D \geq 2.5$ mm), U has a negligible effect on σ compared with that of ε, especially under strongly turbulent conditions ($\varepsilon \geq \varepsilon_{60\%}$).

3.4. Raindrop Axis Ratios

The axis ratio is used to characterize the shape of raindrops bearing the assumption of oblate spheroids (Beard & Chuang, 1987). The distributions of the axis ratio with D under different turbulent conditions are shown in Figures 2a–2c. Below moderate turbulence ($\varepsilon < \varepsilon_{60\%}$), the distributions of the mean axis ratio with diameter are similar (Figures 2a and 2b). For raindrops less than 2.5 mm, the axis ratio is greater than that reported in Thurai

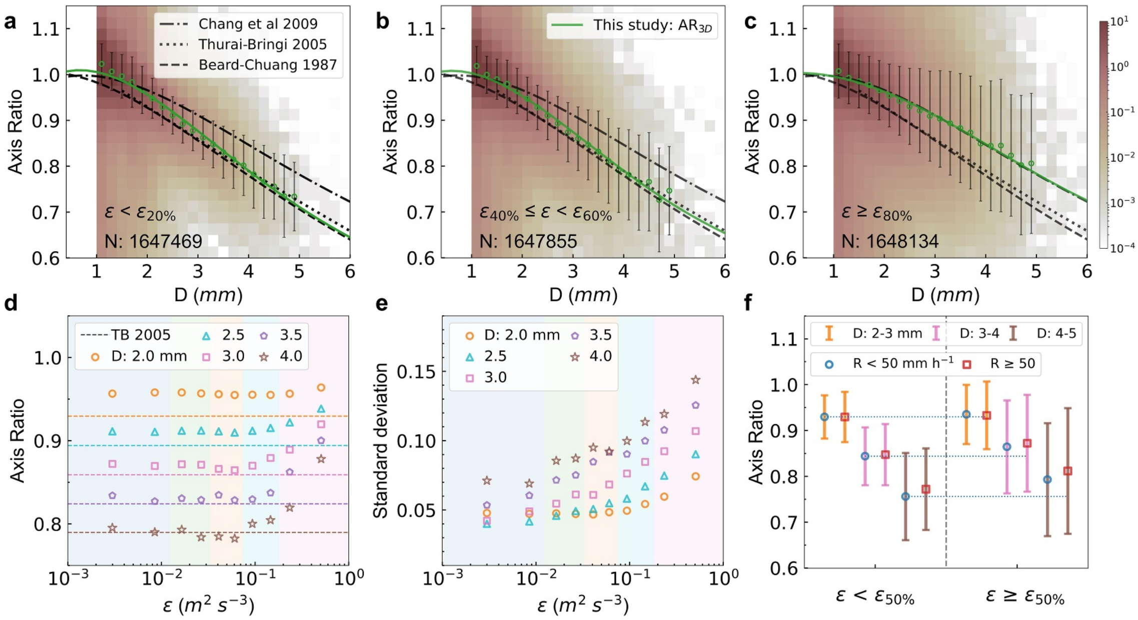

Figure 2. Effect of turbulence on raindrop axis ratios. (a–c) Distribution of raindrop axis ratios under different ε, where the green solid line represents the fitting result of this study using a polynomial, the green hollow dots represent the mean axis ratios of the diameter bins, and the error bars represent the standard deviations. The dashed-dotted line, dotted line, and dashed line represent the results of Beard and Chuang (1987), Chang et al. (2009), and Thurai and Bringi (2005), respectively. The fitting equations are given in Table S3 in Supporting Information S1. (d, e) Distribution of the mean and standard deviation results of the axis ratio with increasing ε for different diameters ($D \pm 0.1$ mm) (ε bin: 10%). (f) Distribution of the mean axis ratio and standard deviation under different rain rates ($R < 50$ mm hr^{-1} and $R \geq 50$ mm hr^{-1}) and different ε values ($\varepsilon < \varepsilon_{50\%}$ and $\varepsilon \geq \varepsilon_{50\%}$; $\varepsilon_{50\%} = 0.050$ m^2 s^{-3}).

and Bringi (2005) (artificial raindrops falling from a bridge 80 m above ground observed by a 2DVD under calm conditions) and (Beard & Chuang, 1987) (theoretical equilibrium shape model). For raindrops larger than 2.5 mm, there is better agreement among the three. This result indicates that for large natural raindrops larger than 2.5 mm, the axis ratio hardly varies below moderate turbulent conditions ($\varepsilon < \varepsilon_{60\%}$) and is slightly greater than that of the theoretical model of Beard and Chuang (1987).

However, under very strong turbulent conditions ($\varepsilon \geq \varepsilon_{80\%}$), the raindrop axis ratio is significantly greater than those reported by Beard and Chuang (1987) and Thurai and Bringi (2005), which are similar to those observed in typhoon passages (Chang et al., 2009). Moreover, the standard deviation of the raindrop axis ratio is significantly larger under very strong turbulent conditions, indicating enhanced raindrop oscillation.

The mean and standard deviation results of raindrop axis ratios with increasing ε for different diameters ($D \pm 0.1$ mm) are given in Figures 2d and 2e, where the filled colors correspond to the five bins of ε in Figure 1a. When $\varepsilon < \varepsilon_{60\%}$, the mean values of the raindrop axis ratio of each D segment change slightly with increasing ε. When $\varepsilon \geq \varepsilon_{60\%}$, the raindrop axis ratios increase significantly, particularly for larger raindrops ($D \geq 2.5$ mm), indicating deviation from the equilibrium shape and a tendency to become rounded.

The differential reflectivity of polarimetric radar is correlated with σ and the axis ratio of raindrops within the sampling volume. This study shows that both σ (as shown in Figure 1d) and the axis ratio (as shown in Figure 2d) of large raindrops ($D \geq 2.5$ mm) increase significantly under strong turbulence. This could severely impact the actual observation of the polarimetric radar variables; that is, a decrease in differential reflectivity under strongly turbulent conditions is expected (Li et al., 2018).

The oscillation amplitudes, characterized by the standard deviation of the raindrop axis ratio (error bars in Figures 2a–2c) (Szakáll et al., 2010), increase with increasing D and ε (as shown in Figure 2e). Figure S3 in Supporting Information S1 also shows that the distribution of the axis ratios broadens significantly with increasing D and ε. Large raindrops exhibit a hysteresis in speed due to their inertia (Zhu, Kollias, & Yang, 2023); however, our results indicate that large raindrops are more likely to trigger high oscillation amplitudes, especially

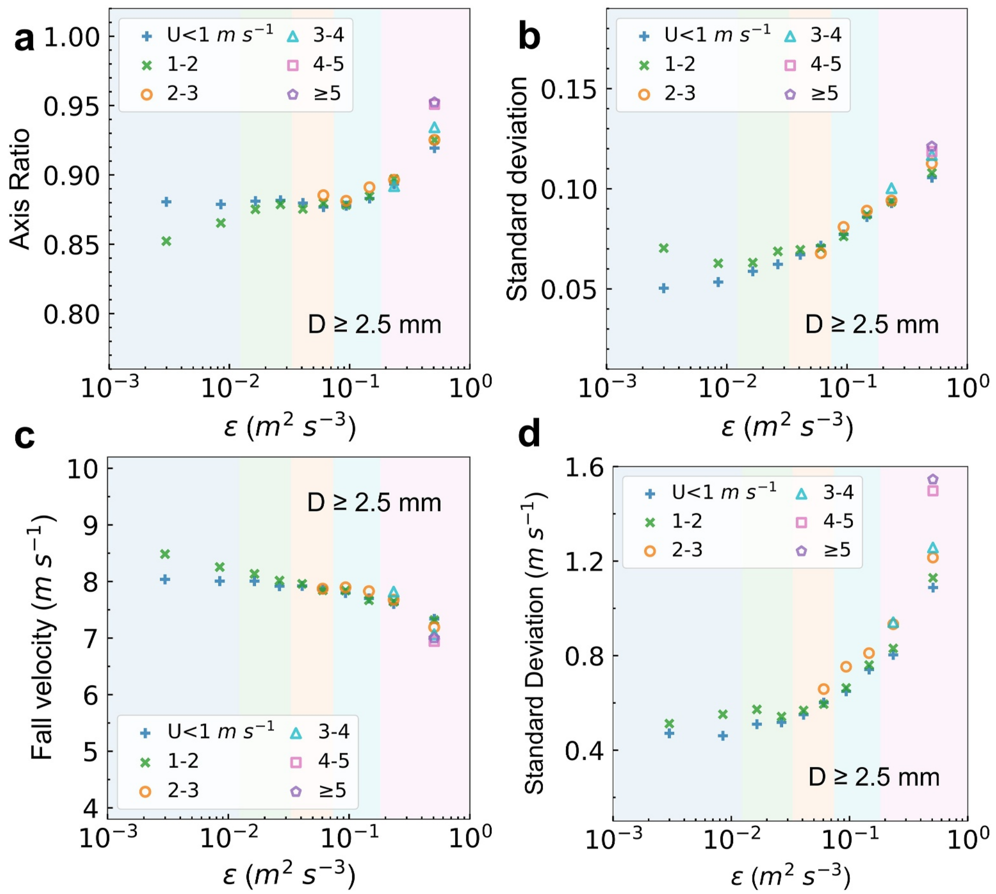

Figure 3. Effect of U on raindrop axis ratios and fall speeds. (a, b) Distribution of the mean axis ratio and standard deviation with ε for $D \geq 2.5$ mm at different U (Note: only the mean axis ratio and standard deviation with a raindrop number of ≥500 are shown). (c, d) The same as in (a) and (b) but for the fall speed.

under more turbulent conditions. This finding may be partially explained by the decreasing effects of surface tension in resisting drop oscillations, which needs further study.

The concentration of raindrops tends to increase as R increases (Wen et al., 2018), which would increase the likelihood of collision and triggers oscillation. Beard (1984) argued that the mean raindrop axis ratio is greater under a stronger R based on the energy perspective. However, recent observations have shown that compared with wind, R has a minimal effect on raindrop axis ratios (Chang et al., 2009). Figure 2f illustrates the distribution of the raindrop axis ratio under weak (R < 50 mm hr^{-1}) and strong ($R \geq 50$ mm hr^{-1}) rain rates for two turbulent conditions ($\varepsilon < \varepsilon_{50\%}$ and $\varepsilon \geq \varepsilon_{50\%}$). The mean raindrop axis ratios are greater for stronger R values. Notably, turbulence enhancement is a more important factor for increasing raindrop axis ratios and oscillation amplitudes.

Similar to Figure 1f, Figure 3 shows the effect of wind on the mean and standard deviation of the raindrop axis ratios and fall speeds. The results show that for large raindrops ($D \geq 2.5$ mm), compared with ε, U has a negligible effect, except in cases of extreme turbulent conditions ($\varepsilon \geq \varepsilon_{90\%}$). Under these conditions, an increase in U tends to increase both the mean and standard deviation of the axis ratio but decreases the mean fall speed. Notably, an increase in U leads to a significant increase in the standard deviation of the fall speed.

3.5. Falling Speeds of Raindrops

Figures 4a–4c show the distributions of raindrop fall speeds with D under different ε. Similar to the axis ratios (Figures 2a and 2b), when $\varepsilon < \varepsilon_{60\%}$, the mean raindrop fall speeds are close to the laboratory terminal fall speed (V_t) (Atlas et al., 1973). Under very strong turbulent conditions ($\varepsilon \geq \varepsilon_{80\%}$), the mean fall speeds of raindrops with

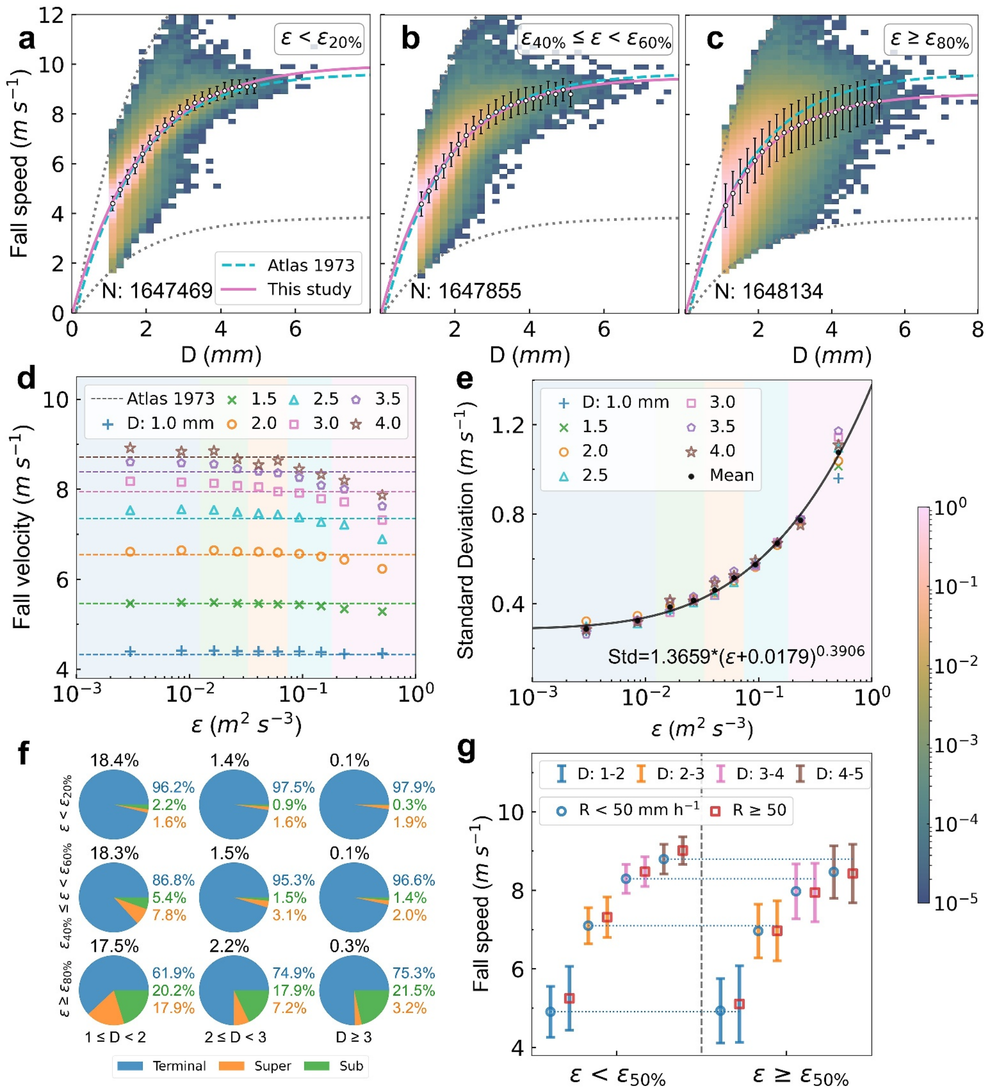

Figure 4. Effect of turbulence on raindrop fall speed. (a–c) Distribution of raindrop fall speed under different ε, where the pink solid line represents the fitting result of this study, the white hollow dots represent the mean fall speed for each diameter bin, and the error bars represent the standard deviation. The cyan dashed line represents the results of Atlas et al. (1973). The fitting equations are given in Table S4 in Supporting Information S1. (d, e) Distribution of the mean and standard deviation results of the fall speed with increasing ε for different diameters ($D \pm 0.1$ mm) (ε bin: 10%). (f) Percentage of raindrops with fall speeds at Terminal (V_t), Super ($\ge 0.85V_t$), and Sub ($<0.85V_t$) under different D and ε values. (g) Distribution of the mean fall speed and standard deviation under different rain rates ($R < 50$ mm hr^{-1} and $R \ge 50$ mm hr^{-1}) and different ε values ($\varepsilon < \varepsilon_{50\%}$ and $\varepsilon \ge \varepsilon_{50\%}$; $\varepsilon_{50\%} = 0.050$ m^2 s^{-3}).

diameters larger than 2.5 mm are lower than V_t. The statistical results confirmed the presence of sub-terminal raindrops, which have been reported in recent case studies (Bolek & Testik, 2021; Bringi et al., 2018; Testik & Bolek, 2023; Thurai et al., 2019).

The means and standard deviations of the fall speeds of raindrops with different D values as affected by ε are shown in Figures 4d and 4e. For small raindrops ($D < 1.5$ mm), their fall speeds are close to V_t and are barely affected by ε. With increasing ε, the fall speed of large raindrops decreases significantly. The critical value of ε at which this decreasing trend starts to appear is closely related to the diameter of the raindrops. Small raindrops may be more prone to relevant forcings given their relatively small inertia (Testik & Bolek, 2023). Interestingly, the critical value of ε is lower for larger raindrops (Figure 4d), indicating that the impact of turbulence on raindrop fall speeds becomes more significant as D increases.

Notably, raindrop fall speeds for a specific diameter follow the Gaussian distribution well (as shown in Figure S4 in Supporting Information S1), and the standard deviation of raindrop fall speeds does not vary much with D at the same ε, except under extreme turbulent conditions ($\varepsilon \geq \varepsilon_{90\%}$) (Figure 4e and Figure S4 in Supporting Information S1). The standard deviation gradually increases with increasing ε in all raindrop diameter bins. The study also provides a quantitative fitting relationship, as shown in Figure 4e: $Std = 1.3659(\varepsilon + 0.0179)^{0.3906}$.

Figure 4f shows the percentages of raindrops falling at terminal (V_t), sub-terminal ($<0.85V_t$) and super-terminal ($\geq 1.15V_t$) speeds (Testik & Bolek, 2023). Based on observations during 17 different rainfall events, Testik and Bolek (2023) reported that the percentage of sub-terminal (super-terminal) raindrops is 18.5% (9.5%). However, our results show that most raindrops fall at V_t under moderate or less turbulent conditions ($\varepsilon < \varepsilon_{60\%}$). Only under very strong turbulent conditions ($\varepsilon \geq \varepsilon_{80\%}$), the percentage of sub-terminal raindrops increases rapidly, up to approximately 20% for all D ranges. Notably, the percentage of super-terminal raindrops is remarkable (17.9%) in only the small D range (D: 1–2 mm) under very strong turbulent conditions, which may be related to the maintenance of fall speed after raindrop breakup or preferential sweeping (Larsen et al., 2014; Montero-Martínez et al., 2009).

The impact of R is given in Figure 4g. The results show that turbulence enhancement is a more dominant factor for increasing the standard deviation of raindrop fall speeds than is R enhancement, especially for large raindrops ($D \geq 2$ mm). Notably, compared to lighter rainfall ($R < 50$ mm hr^{-1}), the effect of turbulence enhancement on reducing raindrop fall speeds is more pronounced under strong R conditions ($R \geq 50$ mm hr^{-1}).

4. Conclusions and Implications

In this work, 3 years of in situ observations were used to quantify the impact of turbulence on raindrop microphysics, including raindrop canting angles, shapes and fall speeds. The 3D oblate spheroidal reconstruction technique was applied to 2D shadows of raindrops collected from orthogonal 2D video disdrometer (2DVD) observations, and more than 8 million raindrops were analyzed. The turbulence dissipation rate (ε) was estimated based on data from nearby 3D ultrasonic anemometers. We showed that as the air becomes more turbulent, the physical states of raindrops increasingly deviate from the equilibrium state, and this deviation is pronounced for large raindrops.

We found that turbulence has detectable impact on the range of raindrop canting angles, as quantified by σ. Similar to Huang et al. (2008), σ decreases with D under weakly turbulent conditions. In contrast, under strongly turbulent conditions, σ increases with increasing D and those trends are consistent across different R ranges. In addition, for a given range of D, σ and the logarithm of ε are well positively correlated.

The raindrop oscillation amplitude (characterized by the standard deviation of the raindrop axis ratio) increases with ε, and large raindrops are more likely to trigger high oscillation amplitudes, especially under more turbulent conditions. The mean raindrop axis ratio is minimally affected by weak turbulence ($\varepsilon < 0.1$ m^2 s^{-3}). With a further increase in turbulence, the large raindrops ($D \geq 2.5$ mm) deviated from the equilibrium shape and became more rounded, as evidenced by a significant increase in their mean axis ratio. Compared with an increase in R, enhanced turbulence is a more critical factor in elevating the raindrop axis ratio and oscillation amplitude.

Reduced raindrop fall speeds deviating from the equilibrium state under turbulent conditions were confirmed in this study. We showed that turbulence has a greater impact on the mean fall speeds of larger raindrops, while smaller raindrops are relatively immune. Under very strong turbulent conditions ($\varepsilon \geq \varepsilon_{80\%}$), sub-terminal raindrops (approximately 20%) occurred at all diameters, while super-terminal raindrops were mainly concentrated at small diameters (D: 1–2 mm) (approximately 18%). Notably, the standard deviation of the raindrop fall speed does not vary much with D and R but increases rapidly with increasing ε.

The effects of U on σ, the axis ratio and the fall speed were carefully evaluated, and we concluded that for large raindrops ($D \geq 2.5$ mm), the wind effects were negligible compared with those of ε, except in cases of extreme under extremely turbulent conditions. The ε derived from the 3D ultrasonic anemometer data near the 2DVD could effectively characterize the ε near the raindrops in the core observation area of the 2DVD. In the future, the results can be compared against open measurements made with instruments such as the Video In Situ Snowfall Sensor (Maahn et al., 2023).

The atmosphere is not always heavily turbulent (as shown in Figure 1a); therefore, the adequacy of employing the equilibrium approximation is sound in general. We argue that the impact of turbulence may be considered in extreme conditions such as those that arise during typhoons, extreme rainfall, tornadoes, etc. For example, Li et al. (2023) hypothesized that the underestimation of extreme rainfall is attributed to the assumed raindrop equilibrium state. In numerical models, raindrop coalescence and breakup remain the most uncertain and theoretically challenging liquid microphysical processes to quantify (Morrison et al., 2020). In the pursuit of advancing physically based parameterizations of raindrop collision-coalescence and breakup, it is essential to account for the influence of turbulence on raindrop microphysics (Liu et al., 2023; Morrison et al., 2020). In addition, raindrops are assumed to fall at terminal speeds without a broad distribution of fall speeds affected by turbulence, as observed in this study, even in state-of-the-art Lagrangian particle-based schemes (Shima et al., 2020). Future studies are warranted to represent the interaction between turbulence dynamics and raindrop microphysics in cloud microphysics schemes.

Data Availability Statement

The data that support the analysis and conclusions of this paper are available at Figshare (Zheng, 2024).

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grants 42075080, 42305087 and 41975066), the Basic Research Fund of CAMS (Grant 2022Y008), and the S & T Development Fund of CAMS (Grant 2023KJ047). Dr. Wupeng Xiao and Dr. Xingtao Huang are acknowledged for their fruitful discussions. Dr. Chao Wang is acknowledged for his help in observation maintenance and data processing. We also thank the reviewers for their constructive comments.

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